Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-K Infectious Vascular Diseases

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-18-11-C01				
NPLEX-P1-CARD-18-11-C02				
NPLEX-P1-CARD-18-11-C03				
NPLEX-P1-CARD-18-11-C04				
NPLEX-P1-CARD-18-11-C05				
NPLEX-P1-CARD-18-11-C06				
NPLEX-P1-CARD-18-11-C07				
NPLEX-P1-CARD-18-11-C08				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-11-C01-BD1

Build a vegetation from a normal valve

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-K Infectious Vascular Diseases
Competency: NPLEX-P1-CARD-18-11-C01 Activity: NPLEX-P1-CARD-18-11-C01-BD1 Date: _____

Without looking at your notes, draw a heart valve in cross section and build the six-step sequence that turns it into an infected vegetation, one step per line and in order. Beside the first step write what kinds of flow and what foreign material injure valve endothelium, and what is exposed underneath once it is injured. Beside the sterile deposit step write what the deposit is made of and what it is called. Beside the adhesion step write the molecule viridans streptococci use to stick to it and the two host proteins that Staphylococcus aureus surface proteins bind. Then write why neutrophils clear this mass poorly and what that does to the blood cultures. Finish by writing which organism can skip the first three steps and infect a valve that was never damaged.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-11-C01-BD2

Reason from the setting to the organism

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-K Infectious Vascular Diseases
Competency: NPLEX-P1-CARD-18-11-C01 Activity: NPLEX-P1-CARD-18-11-C01-BD2 Date: _____

Six patients have infective endocarditis. Patient 1 has had 6 weeks of low-grade fever and weight loss after a tooth extraction, with known mitral regurgitation. Patient 2 injects drugs and has 4 days of high fever with a new holosystolic murmur at the left lower sternal border. Patient 3 had a prosthetic aortic valve placed 3 weeks ago. Patient 4 is 78 and became febrile after a urologic procedure. Patient 5 has bacteremia and iron deficiency anemia with blood in the stool. Patient 6 has 2 months of fever and a vegetation on echocardiography with three sets of negative blood cultures. For each patient name the organism you expect, write the route by which it entered the blood, and write in one line the property of the organism or the surface that makes it the expected one. Then write which patients you expect to have an acute course and which a subacute course, and say what decides that.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-11-C01-RT01

Rebuild the vegetation sequence and mark where vegetations sit

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-K Infectious Vascular Diseases
Competency: NPLEX-P1-CARD-18-11-C01 Activity: NPLEX-P1-CARD-18-11-C01-RT01 Date: _____

Rebuild the six-step formation of a vegetation from memory, numbering each step. Then draw the mitral and aortic valves and mark with an arrow the surface of each one where a vegetation forms. Beside each arrow write whether that surface faces the higher or the lower pressure chamber and why the regurgitant jet lands there. Write which two valves are infected most often and why, then write the one setting in which the tricuspid valve is the most common site and why the organisms reach it first.

Rebuild from memory: Formation of a vegetation

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-18-11-C01-Q01

Organism in right-sided endocarditis

A 29-year-old man who injects heroin comes to the emergency department because of 4 days of fever and shortness of breath. Temperature is 39.4 °C (102.9 °F), pulse is 112/min, respirations are 24/min, and blood pressure is 104/62 mm Hg. Cardiac examination shows a holosystolic murmur at the left lower sternal border that increases with inspiration. A chest radiograph shows multiple peripheral nodular opacities, two of which are cavitating. Which of the following organisms is most likely to be isolated from his blood?

A. Streptococcus gallolyticus

B. Enterococcus faecalis

C. Staphylococcus aureus

D. Staphylococcus epidermidis

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-11-C02-BD1

Build the complication table by mechanism

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-K Infectious Vascular Diseases
Competency: NPLEX-P1-CARD-18-11-C02 Activity: NPLEX-P1-CARD-18-11-C02-BD1 Date: _____

On blank paper build a table of the complications of bacterial endocarditis with one row for each of these mechanisms: valve destruction, perivalvular abscess, left-sided emboli, right-sided emboli, immune complexes, and arterial wall infection. Use the columns Findings and Clue in a question, and fill every cell. Under the abscess row write which valves it complicates most often and name the two conduction structures that lie next to the aortic annulus. Under the immune complex row write which type of hypersensitivity this is, what the urine shows, and what happens to complement levels. Finish by naming the leading cause of death in endocarditis and writing in two sentences how valve destruction produces it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-11-C02-BD2

Sort four new findings back to their mechanisms

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-K Infectious Vascular Diseases
Competency: NPLEX-P1-CARD-18-11-C02 Activity: NPLEX-P1-CARD-18-11-C02-BD2 Date: _____

Four patients already known to have endocarditis develop something new. Patient 1 has a splenic infarct and a left hemiparesis. Patient 2 has multiple cavitating lung nodules with pleuritic pain and hemoptysis. Patient 3 has a PR interval that has lengthened from 0.16 s to 0.28 s over two days, with an infected aortic valve. Patient 4 has hematuria with red cell casts, a low serum complement level, and painful nodules on the finger pads. For each patient name the mechanism, name the infected valve or the site of the new lesion, and explain in two or three sentences how the mechanism produces that exact finding. Then write what sudden severe headache in week 6 of the illness suggests and explain how a vegetation fragment weakens an arterial wall. Finish by writing why sudden severe regurgitation causes pulmonary edema while regurgitation that develops over years does not.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-18-11-C02-CA01

Six weeks of fever, blood in the urine, and a changing murmur

A 64-year-old woman comes to the physician because of 6 weeks of low-grade fever, night sweats, fatigue, and a 4.5 kg (10 lb) weight loss. She had a tooth extracted 2 months ago. She has had a murmur of mitral regurgitation for 12 years. Temperature is 38.1 °C (100.6 °F), pulse is 96/min, respirations are 16/min, and blood pressure is 128/76 mm Hg. Cardiac examination shows a holosystolic murmur at the apex that is louder than it was recorded 1 year ago. There are two tender nodules on the pads of the left index and ring fingers and painless flat red macules on both palms. Funduscopic examination shows two retinal hemorrhages with pale centers. Laboratory studies show a hemoglobin of 10.1 g/dL (reference range 12.0 to 15.5 g/dL), erythrocyte sedimentation rate of 72 mm/h (reference range 0 to 20 mm/h), serum creatinine of 1.6 mg/dL (reference range 0.6 to 1.1 mg/dL), and serum C3 and C4 below the reference range. Urinalysis shows 3+ blood with dysmorphic erythrocytes and erythrocyte casts, and no bacteria.

1. Recognize. Name the infection, the valve involved, the course (acute or subacute), and the organism you expect from this history, and name each of the four peripheral findings described.

2. Explain. Explain how the organism came to sit on this particular valve, writing the sequence from the dental extraction to a growing vegetation and saying what the regurgitant jet contributed.

3. Differentiate. Write which of her findings come from embolism and which come from immune complexes, and explain what separates the two mechanisms in this patient.

4. Investigate. Write what the urinary findings, the dysmorphic erythrocytes, and the low C3 and C4 together tell you about the kidney lesion, and name the hypersensitivity mechanism and the structure in which the complexes deposit.

5. Decide. State which single new finding over the next week would tell you the infection has extended into the aortic annulus, and state which finding would tell you a leaflet or chord has given way.

6. Justify. Justify both of those predictions from anatomy: name the conduction structures next to the aortic annulus, and explain why a ventricle facing sudden severe regurgitation develops pulmonary edema.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-11-C03-BD1

Trypanosoma cruzi from the bug to the tissue

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-K Infectious Vascular Diseases
Competency: NPLEX-P1-CARD-18-11-C03 Activity: NPLEX-P1-CARD-18-11-C03-BD1 Date: _____

Without looking at your notes, write out how Trypanosoma cruzi gets into a person and what it does once it is inside. Name the organism type, the geographic region, the vector and where it lives, when it feeds, and the exact step by which infective forms enter the body. List the four other routes of transmission. Then write the two forms of the parasite in human tissue, which one lives inside cells and multiplies, and which one travels in the blood. Draw the acute entry findings: label the skin lesion and the eye finding by name and write which entry site produces each. Finish by writing how long the acute phase lasts, how the parasite is found in the blood during it, and who is at risk of acute myocarditis or meningoencephalitis.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-11-C03-BD2

Explain each sign of chronic Chagas disease

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-K Infectious Vascular Diseases
Competency: NPLEX-P1-CARD-18-11-C03 Activity: NPLEX-P1-CARD-18-11-C03-BD2 Date: _____

A 52-year-old man who moved from rural Bolivia 20 years ago has an enlarged heart with a low ejection fraction, right bundle branch block with left anterior fascicular block, a thin bulging left ventricular apex, dysphagia, and severe constipation. For each of those findings write what has been destroyed in the tissue and how that destruction produces the finding. Write where the parasite has persisted and what the heart tissue looks like under the microscope, including how many parasites you expect to see. Explain why sudden death can be the first sign, naming the mechanism in the scarred myocardium, and explain what forms in the apical aneurysm and where it travels. Then write why a blood smear is of no use now while it worked in the acute phase, and name what is used instead.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-18-11-C03-CP01

Acute against chronic Chagas disease

Fill in the table from memory, then check it against the notes and correct any cell you missed.

Feature	Acute	Chronic
Timing after infection and the share of people affected		
Parasite numbers and where the parasites are		
Main findings outside the heart		
Heart findings		
How the infection is detected		
What the tissue looks like under the microscope		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-11-C04-BD1

Build the three stages of Lyme disease

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-K Infectious Vascular Diseases
Competency: NPLEX-P1-CARD-18-11-C04 Activity: NPLEX-P1-CARD-18-11-C04-BD1 Date: _____

Name the organism that causes Lyme disease and the kind of bacterium it is, then name the tick genus and the two species, the animal reservoir, the animal that sustains the adult ticks, the regions of the United States where risk is concentrated, and the season when most infections are acquired. Write which stage of the tick transmits most infections and why it is missed. Write how long the tick must stay attached and explain that number from where the spirochete sits in the tick. Then build a table of the three stages with one row each for early localized, early disseminated, and late disease, using the columns Timing, Findings, and Mechanism, and fill every cell. Describe the appearance and the size of the early skin lesion, and name the two other infections the same tick can transmit.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-11-C04-BD2

Lyme carditis and why the testing changes with the stage

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-K Infectious Vascular Diseases
Competency: NPLEX-P1-CARD-18-11-C04 Activity: NPLEX-P1-CARD-18-11-C04-BD2 Date: _____

A 19-year-old from rural Wisconsin had an expanding red rash on the thigh 3 weeks ago and now has lightheadedness, a pulse of 40/min, and complete heart block with a narrow QRS. Write which stage of Lyme disease this is, which tissue is inflamed and what the infiltrate is made of, and why the degree of block can change within hours. Write whether you expect the block to resolve and explain your reasoning from what is happening in the node, inflammation or scarring. Then write why a patient with the early skin lesion alone is recognized clinically rather than by antibody testing, and what testing is used once the infection has disseminated. Finish by writing why bilateral facial palsy points toward Lyme disease, and name two other causes of inflamed or destroyed conduction tissue from this chapter and the feature that separates each one from Lyme carditis.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-18-11-C04-Q02

Heart block after a summer rash

A 19-year-old woman comes to the emergency department in August because of 2 days of lightheadedness. She lives in rural Wisconsin and spends weekends hiking. Three weeks ago she had an expanding red patch on the posterior thigh that was 9 cm across and cleared in the center. Temperature is 37.4 °C (99.3 °F), pulse is 40/min, respirations are 14/min, and blood pressure is 102/64 mm Hg. The electrocardiogram shows P waves at 82/min that are unrelated to QRS complexes at 40/min with a duration of 0.08 s. Examination shows weakness of the right side of the face that includes the forehead. Which of the following best explains the conduction abnormality in this patient?

A. Lymphocytic inflammation of the atrioventricular node around a disseminated spirochete

B. Fibrosis of the conduction system after decades of intracellular protozoal persistence

C. Extension of valvular infection into the aortic annulus as a ring abscess

D. Deposition of circulating immune complexes in the conduction system with complement activation

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-11-C05-BD1

Rickettsia rickettsii from the tick to the vessel wall

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-K Infectious Vascular Diseases
Competency: NPLEX-P1-CARD-18-11-C05 Activity: NPLEX-P1-CARD-18-11-C05-BD1 Date: _____

Name the organism that causes Rocky Mountain spotted fever and describe what kind of bacterium it is, including why it stains poorly on Gram stain and where it must live to replicate. Name the tick genus and the two United States tick species, name the states where most cases occur, give the season, and give the number of days between the bite and the first symptoms. Then write the cell the organism infects and rebuild in order what it does to that cell: how it gets in, how it escapes the phagosome, what it polymerizes, and how it reaches the next cell. Write the name of the tissue lesion this produces and list the six organs the notes say it reaches. Finish by drawing the rash: write the day it appears, where it starts, the direction it spreads, what it looks like at first, what it becomes, and which body surfaces are characteristically involved late.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-11-C05-BD2

Derive every laboratory finding from the leaking vessel

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-K Infectious Vascular Diseases
Competency: NPLEX-P1-CARD-18-11-C05 Activity: NPLEX-P1-CARD-18-11-C05-BD2 Date: _____

Starting from injured endothelium alone, explain each of these findings in two or three sentences: thrombocytopenia, hypoalbuminemia with edema, hyponatremia, a petechial rash, noncardiogenic pulmonary edema, confusion and seizures, and shock with acute kidney injury. For the low platelet count say whether the marrow is the problem and name where the platelets are going. For the low sodium name the hormone involved and the stimulus that releases it. Then write the classic triad, write how often all three are present at the start, and explain why the diagnosis is made clinically in the first week rather than on antibody testing. Finish with a 44-year-old in North Carolina in May who has fever, headache, vomiting, and abdominal pain and no rash: write which three other causes of a rash on the palms and soles you would keep in mind and the single feature that separates each from this illness.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-18-11-C05-PR01

Predict what happens as the vasculitis spreads

A 38-year-old man was bitten by a Dermacentor tick in Oklahoma in June and now has his fifth day of fever and headache. Rickettsiae are multiplying in the endothelium of small vessels in his skin, lungs, brain, kidneys, and gut. Predict each item below, write the direction of the change, and give in one line the vascular event that explains it.

What changes	Up, down, or no change	Why, in one or two lines
What happens to the permeability of the small vessels and to the fluid in the interstitium		
What happens to the serum albumin and to the effective circulating volume		
What happens to ADH release and to the serum sodium		
What happens to the platelet count, and whether the marrow or the vessel wall accounts for it		
What happens to the skin lesions as red cells escape, and where they appear first		
What happens to oxygenation when the same process occurs in the pulmonary capillaries		
What happens to blood pressure and urine output as the leak continues		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-18-11-C05-Q03

Low sodium and low platelets in a spotted fever

A 41-year-old woman comes to the emergency department in May because of 5 days of fever, severe headache, and vomiting. She removed a tick from her calf 1 week ago while gardening in North Carolina. Temperature is 39.6 °C (103.3 °F), pulse is 118/min, respirations are 22/min, and blood pressure is 92/58 mm Hg. Examination shows petechiae on the wrists, ankles, and trunk and pitting edema of both legs. Laboratory studies show a platelet count of 68,000/microliter (reference range 150,000 to 400,000/microliter), serum sodium of 127 mEq/L (reference range 135 to 145 mEq/L), and serum albumin of 2.4 g/dL (reference range 3.5 to 5.0 g/dL). Which of the following best explains this combination of findings?

A. Deposition of circulating immune complexes in small vessels with complement activation

B. Replication of the organism within platelets and marrow megakaryocytes

C. Non-neutralizing antibody increasing viral entry into monocytes and macrophages

D. Infection of vascular endothelium causing plasma leak and platelet consumption at the vessel wall

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-11-C06-BD1

Yellow fever from the mosquito to the liver

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-K Infectious Vascular Diseases
Competency: NPLEX-P1-CARD-18-11-C06 Activity: NPLEX-P1-CARD-18-11-C06-BD1 Date: _____

Write the virus family and the genome type shared by yellow fever and dengue, and name the mosquito that carries both and where it breeds. For yellow fever write the regions where it occurs, the two transmission cycles and the animals in the jungle cycle, the incubation period, and the findings of the first febrile illness. Write the share of patients who improve briefly and then enter the toxic phase. Then draw a liver lobule, mark the zone the virus damages, and write the name of that pattern of necrosis and why it is unusual compared with toxic and ischemic injury. Name the dying hepatocyte by its eponym and describe what it looks like, and write what else the notes say the liver shows. Finish by writing how liver failure produces jaundice and bleeding, explaining the dark vomit, and naming the two organ failures that follow.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-11-C06-BD2

Why the second dengue infection is the dangerous one

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-K Infectious Vascular Diseases
Competency: NPLEX-P1-CARD-18-11-C06 Activity: NPLEX-P1-CARD-18-11-C06-BD2 Date: _____

A woman had dengue three years ago and now has a different serotype. Write how many serotypes exist, then explain step by step why her old antibodies make this infection worse: what the antibodies do to the new virus, which part of the antibody binds which receptor on which cells, and what happens to viral load and cytokine release. Name this mechanism and name the other group of patients at risk through the same route. Then write when in the illness severe disease appears, relative to the fever, and explain what is happening at the capillary wall. Her hematocrit rises from 40% to 48% while her platelets fall to 30,000 per microliter: write what the rising hematocrit means about where her plasma has gone, and explain why it warns of shock before the blood pressure falls. Finish by writing where fluid collects as the leak continues.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-18-11-C06-CP01

Yellow fever against dengue

Fill in the table from memory, then check it against the notes and correct any cell you missed.

Feature	Yellow fever	Dengue
Where in the world it occurs		
Main target cell		
Hallmark pathology		
Key clinical findings		
Laboratory clue		
What the severe form is linked to		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-11-C07-BD1

The three shared effects of hemorrhagic fever viruses

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-K Infectious Vascular Diseases
Competency: NPLEX-P1-CARD-18-11-C07 Activity: NPLEX-P1-CARD-18-11-C07-BD1 Date: _____

Write what viral hemorrhagic fever means and why it is a syndrome rather than one disease. List the four virus families named in the notes with the diseases that belong to each and the genome type of each family. Write what the viruses have in common about their envelope. Then write out the three shared effects that produce the syndrome: for the first, name the cells infected first, where those cells carry the virus, what the viruses do to type I interferon signaling, and what happens to lymphocytes; for the second, name four mediators released and what they do to endothelial junctions; for the third, name the molecule that activated macrophages express, the process it starts, what is consumed, and the other name for that process. Add what hepatocyte injury contributes. Finish by writing why patients leak plasma and bleed at the same time and what most deaths come from.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-11-C07-BD2

Work out who is at risk in a filovirus outbreak

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-K Infectious Vascular Diseases
Competency: NPLEX-P1-CARD-18-11-C07 Activity: NPLEX-P1-CARD-18-11-C07-BD2 Date: _____

Describe the structure and genome of Ebola and Marburg viruses and name the likely reservoir. Write the two ways an outbreak starts in a person and the region where these outbreaks occur, then write exactly how the virus passes from one person to the next and name the two groups at highest risk, explaining what each group contacts. Write when a patient becomes infectious relative to symptoms, what happens to viral load as the illness worsens, and where virus can persist after recovery. State whether ordinary airborne spread occurs. Then give the incubation period and write the findings of the first days and of the second week, and explain which of those findings accounts for most of the fluid and electrolyte loss. Finish by listing the laboratory findings you expect and giving the approximate case fatality.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-11-C07-RT01

Rebuild the common pathway of the hemorrhagic fever viruses

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-K Infectious Vascular Diseases
Competency: NPLEX-P1-CARD-18-11-C07 Activity: NPLEX-P1-CARD-18-11-C07-RT01 Date: _____

Rebuild the seven-step common pathway of the hemorrhagic fever viruses from memory, one step per line and in order, from the first cells infected through to multiorgan failure. Beside the interferon step write what blocking interferon signaling buys the virus. Beside the cytokine step name the mediators. Beside the tissue factor step write which cells express it and name what the consumption of platelets and clotting factors is called. Then mark on your sequence the two points at which the patient loses intravascular volume and the two points at which bleeding becomes likely, and write one sentence on why shock and bleeding appear together rather than one after the other.

Rebuild from memory: Common pathway of hemorrhagic fever viruses

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-11-C08-BD1

Build the course of viral myocarditis

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-K Infectious Vascular Diseases
Competency: NPLEX-P1-CARD-18-11-C08 Activity: NPLEX-P1-CARD-18-11-C08-BD1 Date: _____

Define myocarditis and say what must be excluded as the cause of the myocyte injury. Name the most common group of causes in developed countries, name the classic virus with its virus family, genome and routes of spread, and list the other viruses the notes name. Describe the typical patient and explain why cardiac symptoms appear one to a few weeks after a respiratory or gastrointestinal illness. Then rebuild the six-step course from the first infection to dilated cardiomyopathy, one step per line. Beside the entry step name the receptor on the myocyte and what kind of protein it is. Beside the immune step name the two cell types that destroy infected myocytes and write what cytokines do to contractility. Write what happens in the minority whose inflammation persists, including the cardiac protein autoimmunity can target. Finish by writing which virus infects coronary microvascular endothelium instead of myocytes.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-11-C08-BD2

Separate myocarditis from an infarction and from the other infiltrates

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A 23-year-old man has chest pain, diffuse ST elevation, and a raised troponin one week after a febrile diarrheal illness, and his coronary arteries are normal. Write why the troponin is raised and name the process that is killing the myocytes. Write where the ECG changes, the conduction disturbances, and the arrhythmias come from, and name the population in which this is an important cause of sudden death. Explain why his chest pain is pleuritic and the ST elevation diffuse. Then write what the ejection fraction does as contracting myocytes are lost and the symptoms that follow. Write what an endomyocardial biopsy would show and why a single biopsy can be negative in a patient who has the disease. Write where cardiac MRI shows its abnormality and how that pattern differs from an infarction. Finish by naming the cause suggested by an infiltrate rich in eosinophils and the cause suggested by multinucleated giant cells.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

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Board-style question NPLEX-P1-CARD-18-11-C08-Q04

Biopsy after a viral illness

A 23-year-old man comes to the emergency department because of 2 days of sharp chest pain that is worse when he lies flat. He had 4 days of fever and watery diarrhea 1 week ago. Temperature is 37.8 °C (100.0 °F), pulse is 104/min, respirations are 18/min, and blood pressure is 112/70 mm Hg. The electrocardiogram shows diffuse ST elevation. Serum troponin I is elevated. Coronary angiography shows normal coronary arteries. Echocardiography shows global hypokinesis with an ejection fraction of 38%. Which of the following is most likely to be seen on endomyocardial biopsy?

A. Multinucleated giant cells with extensive myocyte loss

B. A lymphocytic infiltrate with necrosis of adjacent myocytes

C. Intracellular amastigotes with interstitial fibrosis

D. An eosinophil-rich infiltrate with little myocyte necrosis

Why did you pick it?